Do Now · retrieval

Lesson 050 · Queue<T>: FIFO and Service Order

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is stack Pop order?

Q2

What is Peek?

Q3

What does Enqueue do?

Q4

An empty queue executes enqueue(A), enqueue(B), enqueue(C), x=dequeue(), enqueue(x), then dequeue(). Trace the final queue from front to rear.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Observe — A queue is first-in, first-out (FIFO). Enqueue joins the rear; Dequeue removes from the front.

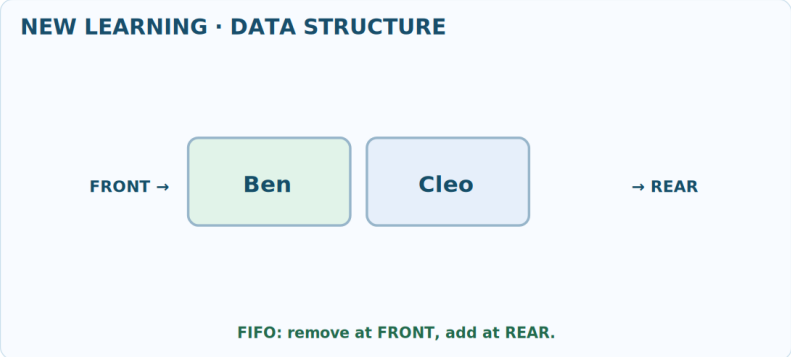
Core idea 2

Explain — Queues are suitable for ordered service requests such as print jobs or breadth-first exploration, if the queueing policy is FIFO.

Core idea 3

Apply — Peek does not remove. Guard Dequeue/Peek with Count>0. Queue<T> handles storage internally; an explicit circular array queue instead tracks front, rear and count and wraps indexes modulo capacity (revisited in Lesson 197).

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

FIFO — First In First Out.

Enqueue — Add at queue rear.

Dequeue — Remove from queue front.

front — Next item to be served.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Trace a ticket queue with Jo then Kai; dequeue one.

Follow the worked method

1. Trace and justify — Jo is served first; Kai remains at the front.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Trace and justify
Dequeue returns 4 then 7. Remaining front-to-back [2,9].

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Queue A,B,C; Dequeue once, then Peek. State removed and front value. [6]

Question 2

Write C# code to enqueue 2,4,6 into Queue<int>, then safely dequeue one. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Compare Stack<T> and Queue<T> for DFS versus BFS on a graph, explaining removal order. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A queue's Dequeue removes the newest item first.

Use key vocabulary from memory

1. Explain FIFO in your own words.

2. Explain Enqueue in your own words.

3. Explain Dequeue in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.